Tyler Busching

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EDUCATION

SUNY Farmingdale State College **Farmingdale, NY**
Bachelor of Science in Computer Engineering Technology *Expected May 2027*

- Cumulative GPA: 3.61/4.0, Dean's List
- Relevant Coursework: Microprocessors, FPGA Design, Digital Electronics, Electric Circuits, Network Analysis, Data Communications and Networking

Suffolk County Community College **Selden, NY**
Associate in Arts with Distinction *Aug 2022 – May 2024*

- Cumulative GPA: 3.6/4.0, Dean's List

TECHNICAL SKILLS

Programming: C, C++, VHDL, Assembly
Hardware: Oscilloscopes, multimeters, logic analyzers, electronic loads, soldering, PC assembly & repair
Embedded Systems & Communications: PIC and ARM Microcontrollers, Arduino, FPGA, UART, I2C, SPI, TCP/IP
Tools: MPLAB IDE, NI LabWindows/CVI, NI Multisim, Proteus, Arduino IDE, Visual Studio, Vivado

ENGINEERING EXPERIENCE

Stellant Systems, Inc. **Melville, NY**
Embedded Engineering Intern *June 2026 – Present*

- Developed PIC32 firmware for PMBus communication with TDK-Lambda power supplies, implementing command-line control, telemetry acquisition, and verification with external current measurements.
- Investigated FPGA SFP connectivity issues, identified reversed transmit/receive connections, and validated corrected hardware through IBERT loopback testing at 1.25 and 3.125 Gb/s.
- Built and debugged a LabWindows/CVI interface for TCP-connected hardware, improving connection persistence and interface responsiveness.
- Supported migration of an embedded C application to replace obsolete hardware; developed and tested UART communication, TCP command interfaces, nonvolatile storage, and RTC functionality.

LEADERSHIP & TECHNICAL ACTIVITIES

IEEE Student Branch – SUNY Farmingdale State College **Farmingdale, NY**
Executive Board Secretary *August 2026 – Present*

- Help organize soldering and microcontroller workshops, collaborative engineering projects, and student resume & career fair preparation activities.
- Designed Arduino Morse code projects and wrote workshop guides covering UART communication, pushbutton input, and LCD output.
- Automated attendance tracking using Google Apps Script, consolidating meeting responses into a membership roster sheet and flagging missing student identification information.

SELECTED PROJECTS

Wireless Networked Power Supply Management System
Senior Design Project | Team Leader *Fall 2026 – Present*

- Leading a three-person team in designing and implementing a modular system for wireless control, monitoring, and independent verification of multiple custom linear DC power supplies.

Custom PC Building & System Optimization *2018 – Present*

- Built and configured 20+ desktop systems, selecting components for performance, compatibility, and budget; diagnosed boot failures, driver conflicts, network problems, and thermal issues.

ADDITIONAL EXPERIENCE

Bourne Mansion, Timber Point Country Club **Great River, NY**
Catering Captain, Assistant Captain, Server *April 2023 – May 2026*

- Led coordination of 250+ guest banquet events, managing teams of 10+ under strict time constraints.